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Summary

Third-year CSE student who tends to build things that already exist — a DOM reconciler, a steganography detector, a swarm simulation — in order to understand how they work at the implementation level. Comfortable across ML pipelines, full-stack, and systems design. Actively seeking internship opportunities.

Technical Skills

Languages	Python, TypeScript, JavaScript, Java, C
Frontend	React, HTML, CSS, Vite
Backend / API	FastAPI, Node.js, REST APIs
ML / AI	PyTorch, Stable-Baselines3, scikit-learn, NumPy, OpenCV
Tools & Design	Git, Linux, Vitest, Figma, Blender

Projects

AEGIS — Autonomous Drone Swarm Defence Simulation



Python, NumPy, Pygame, Stable-Baselines3, Gymnasium | Team · SIH 2025 · Sep – Nov 2025

- Simulated a 4x4 km contested airspace across 4 scenarios (corner assault, multi-vector, comms-denied, comms-degraded) with 10 friendly and up to 13 hostile drones spanning 3 specialised hostile classes (kamikaze, striker, baiter).
- Designed a fully decentralised 6-stage per-drone pipeline — sense, communicate, score, assign, guide, engage — running every 100 ms with no central coordinator; sensor model uses quadratic detection falloff, a $\pm 120^\circ$ FoV cone, and Gaussian position noise scaling to ~ 24 m error at max range.
- Trained 4 simultaneous class-specific adversarial PPO policies in a co-evolutionary loop over 1,000 generations; validated across 50 runs per scenario tracking neutralisation rate, coverage gap time, and friendly loss rate.

Weave — Renderer-Agnostic Virtual DOM Reconciler



TypeScript, Vite, Vitest | Solo · Feb 2026 – Present

- Built a platform-agnostic Virtual DOM reconciler from scratch in TypeScript with zero library dependencies and a ~ 3 KB bundle; enforces strict import boundaries (core imports nothing, renderer imports core, platforms import both) to guarantee portability.
- Implemented keyed and non-keyed diffing producing a 7-type minimal patch set; renderer maintains a `vnodeId` $\rightarrow$ `realNode` map and delegates all platform I/O to a swappable `HostConfig` interface, enabling retargeting to any host environment.
- Shipped a real-time metrics dashboard built entirely on Weave — tracking FPS, patch counts, update durations, and a live patch heatmap — proving the engine can drive complex stateful UIs with built-in instrumentation.

Scry — Steganography Embedding and Detection Engine



Python, FastAPI, React, scikit-learn | Solo · Jan – Mar 2026

- Built a full-stack steganography tool supporting 4 embedding methods (LSB Replacement, LSB Matching, DWT, DCT) with a universal decoder that attempts all in sequence; deployed on Render with a FastAPI backend serving a React + Vite frontend as static files.
- Implemented 4 parallel statistical detectors (chi-square, RS analysis, histogram, entropy) aggregated with type-aware weights from a Random Forest image classifier — trained on 5 image types to suppress false positives on screenshots and AI-generated images — running in production on every request.
- Trained a second per-method stego detection classifier but deliberately withheld it from the UI pending confidence calibration, avoiding exposing uncalibrated probability scores to users.

Education

B.Tech in Computer Science and Engineering — BP Poddar Institute of Management and Technology

2023 – Present

Class XII (CBSE) — Barasat Indira Gandhi Memorial High School

2021 – 2023

Class X (ICSE) — Auxilium Convent School, Barasat

2009 – 2021

Certifications & Interests

Ethical Hacking from Scratch — Udemy (2024)

Interests: Chess, Football, Cycling, Esports